

Apparent Shape Manipulation by Light-Field Projection onto a Retroreflective Surface

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ABSTRACT

For the correct optical presentation of metallic luster structural color, reproducing the changes in brilliance and color that accompany the movement of the viewpoint is essential. Light-field projection onto a retroreflective surface can optically present a texture depending on the viewpoint. Through this application, the apparent shape can potentially be manipulated depending on the viewpoint. This paper proposes an optical illusion that manipulates the apparent shape of a 3D object using light-field projection based on a perceptual normal map transformation.

Keywords: Retroreflection, Optical illusion

Index Terms: Virtual reality, Light fields, Optical reflection, Cameras, Calibration

1 INTRODUCTION

Many projection mapping techniques, such as dynamic projection mapping, projection onto deformable objects, high-speed projection, multispectral projection, and light-field projection, have been proposed over the past two decades [1, 2]. These methods enable dynamic projection mapping [3], as well as complex projections such as rapidly moving deformable surfaces including the human face [4] and arm [5]. However, typical commercial applications still rely on the assumption of Lambert reflection, which restricts projection mapping to an optical texture mapping in a particular viewing direction.

Light-field projection onto a retroreflective surface enables the viewing of a direction-dependent display, such as structural color [6]. As the specular has disparity for both eyes, it is necessary not only for structural color but also for the shiny surfaces. Furthermore, the optical principle of the light-field projection is the same as that of an autostereoscopic display employing a retroreflective screen [7]. Such light-field projection enables direction-dependent apparent color manipulation [8] and offers potential for the novel projection mapping that alternates the perceptual object shape optically. It should be noted that this is not for a particular viewing direction, but for the entire viewing direction.

This paper proposes a perceptual object bumpiness manipulation using light-field projection onto a retroreflective 3D object using a projector array system. As a demonstration of our method, we present an optical illusion that manipulates an object's shape into physically impossible shapes.

2 PROPOSED METHOD

Figure 1 depicts the flowchart of the proposed method. The purpose of this study is to reproduce bidirectional reflectance distribution function (BRDF) corresponding to the viewpoint

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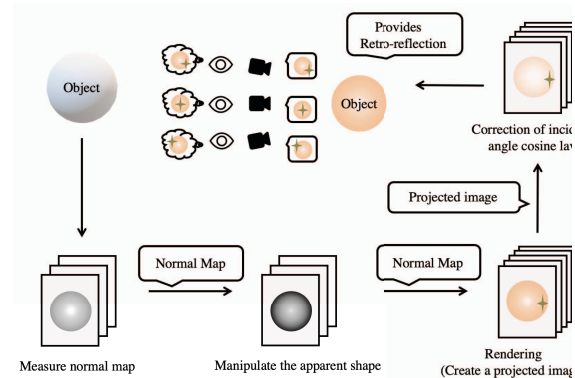


Figure 1: Flow of proposed method.

movement on a manipulated shape for a 3D object with an arbitrary shape. However, to support multi-degree-of-freedom viewpoint movement, it is necessary to project from all directions, namely up, down, left, and right, with respect to the presentation target. In such a case, it is necessary to construct a large-scale system, and the problem of obstructing the viewer's field of view occurs. Therefore, we construct a system in which seven projectors are arranged in the horizontal direction and only support the movement of the viewpoint in the left-right direction. It is thought that this provides the minimum functions because it can reproduce not only the change in gloss and color owing to the movement of the viewpoint, but also the parallax of the specular with both eyes using such movement in the left-right direction.

To generate a projected image that presents a BRDF for any shape, the shape of the object is required. In this research, we obtain a normal map to be presented by using photometric stereo by employing a projector array system and a parallax image by pix-map.

2.1 Apparent shape manipulation and geometric transformation

We manipulate the obtained normal vector to change the apparent shape of the object. Consider an ellipsoid for manipulating normal vectors. First, we calculate a coordinate on the sphere that corresponds to each normal vector, as illustrated in Fig. 2. From the obtained coordinates, the normal vector is obtained from the same (x, y) location on an ellipsoid that is transformed from the sphere. It is possible to adjust the amount of shape manipulation by changing the length of the major axis with the parameter a .

In the normal map obtained in this manner, the apparent shape due to the difference in the viewpoint remains the same as the viewpoint of the camera. Therefore, it is necessary to perform a coordinate transformation to fit each projection viewing position. We convert the normal map into each projector viewpoint using a pixel map that is obtained by gray code projection.

2.2 Creation of projection image

The BRDF sampling produces a projected image for each projector. The BRDF is a function with parameters for the directions of the light source and the direction of the viewpoint. Let θ_{in} be the angle between the light source and the normal, and θ_{out} be the angle between the viewpoint and normal. Using the normal vector $\mathbf{n}$ and light source vector $\mathbf{l}$, the angle of incidence on the surface point can be described as

$$\theta_{in} = \arccos \frac{\mathbf{n} \cdot \mathbf{l}}{\|\mathbf{n}\| \|\mathbf{l}\|}. \quad (1)$$

Moreover, the reflection angle on the same point is

$$\theta_{out} = \arccos \frac{\mathbf{n} \cdot \mathbf{v}}{\|\mathbf{n}\| \|\mathbf{v}\|} \quad (2)$$

where $\mathbf{v}$ is the viewing direction, which is the same as the projection location. It should be noted that the projection that is projected from the projector onto the reflective surface is presented at the position of the projector. φ_{in} and φ_{out} are calculated under the isotropic assumption.

In this study, $\mathbf{n}$ is established by the obtained normal map, and $\mathbf{l}$ is regarded as a parallel light source at the desired position, which is the same as the normal room lighting environment.

2.3 Correction of incident angle cosine law

The desired appearance for each viewing direction can be obtained as described above. However, shading arises because of the angle of incidence cosine law for the Lambert reflection component in the retroreflection surface when these images are simply projected. Therefore, we need to compensate for this effect based on the incident angle cosine law. We solve this problem using the brightness compensation that is calculated from the designed incident angle and obtained normal map.

3 RESULTS

In this study, Michelangelo's David statue and spheres made of plaster were provided with retroreflective properties by spraying. Each normal map was captured using photometric stereo. Furthermore, we obtained a pixel map between the camera and each projector. We subsequently performed rendering using MERL BRDF[9] and projected these images from the corresponding projectors. Figure 3 presents the projection results for each parameter and the distribution of the brightness of the sphere. When $a = -1$, the apparent shape was concave; thus, the movement of the reflected light was opposite to $a = 1$. When $a = 2$, the apparent shape was emphasized.

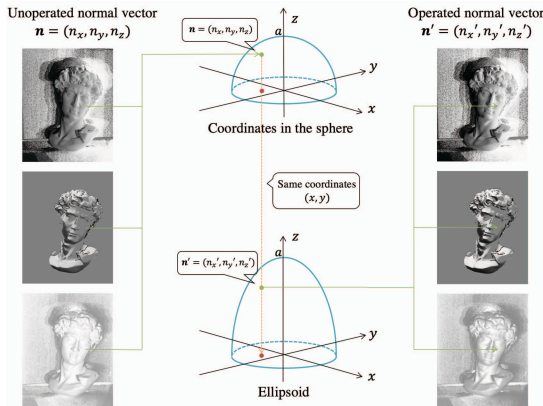


Figure 2: Apparent shape manipulation

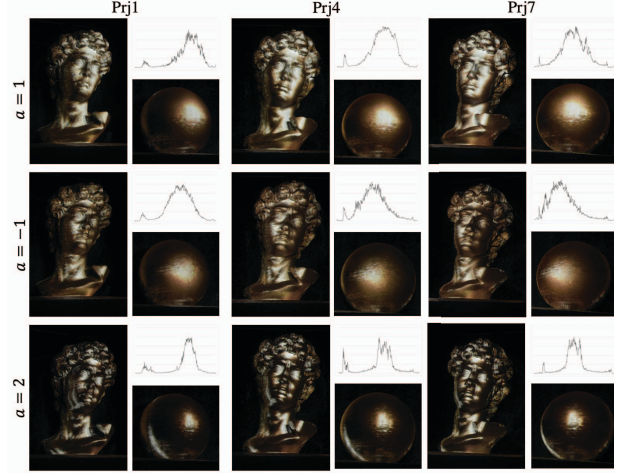


Figure 3: Projection results

4 CONCLUSIONS

In this study, light-field projection was performed on a retroreflective surface to manipulate the apparent shape of the projection target depending on the viewpoint. Through the projection results, we confirmed that the appearance changes depending on the manipulated value. However, although the apparent shape definitely changes, it is incomplete as an illusion that perceives shape change because the gradation due to the cosine law is not perfectly removed. In future work, we would like to evaluate the effectiveness through subject experiments and improve the manipulation quality.

REFERENCES

- [1] O. Bimber and R. Raskar. Spatial Augmented Reality: Merging Real and Virtual Worlds. A. K. Peters, Ltd., 2005.
- [2] A. Grundhöfer and D. Iwai, "Recent Advances in projection mapping algorithms, hardware and applications," *Computer Graphics Forum*, vol. 37, no. 2, pp. 654–675, 2018.
- [3] C. Resch, P. Keitler, and G. Klinker, "Sticky Projections - A Model-Based Approach to Interactive Shader Lamps Tracking," *IEEE Transactions on Visualization and Computer Graphics*, vol. 2626, no. c, pp. 1–1, 2015.
- [4] A. H. Bermano, M. Billeter, D. Iwai, and A. Grundhoefer. Makeup lamps: Live augmentation of human faces via projection. *Computer Graphics Forum*, 36(2):311–323, 2017.
- [5] H. L. Peng and Y. Watanabe, "High-Speed Human Arm Projection Mapping with Skin Deformation," *SIGGRAPH Asia 2020 Emerging Technologies*, SA 2020, 2020.
- [6] T. Amano, K. Minami, "Structural color display on retroreflective objects," In *Proceedings of the 25th International Conference on Artificial Reality and Telexistence and 20th Eurographics Symposium on Virtual Environments*, ICAT – EGVE'15, Eurographics Association, pp. 37–44, 2015.
- [7] K. Nagano *et al.*, "An autostereoscopic projector array optimized for 3D facial display," *ACM SIGGRAPH 2013 Emerging Technologies on - SIGGRAPH '13*, vol. 28, pp. 1–1, 2013.
- [8] T. Amano., S. Ushida. & Y. Miyabayashi. 2017. Viewpoint-Dependent Appearance Manipulation with Multiple Projector-Camera Systems. *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*. 101-107.
- [9] Wojciech Matusik and Hanspeter Pfister and Matt Brand and Leonard McMillan, "A Data-Driven Reflectance Model", *ACM Transactions on Graphics*, vol.22, no.3,759-769,2003.